PROBLEM

What was happening — and why it mattered

WHAT WAS HAPPENING

Case managers review 20–75 documents per claim — intake notes, medical reports, counselling plans, RTW docs — spending 90+ minutes of manual cross-referencing to surface a single referral opportunity.

WHO WAS AFFECTED

Case managers in Healthcare Programs (~70 contracted services across BC); injured workers whose recovery windows close before referrals are recognized; program managers measuring intervention timing.

WHY IT MATTERED

As claim complexity grows, critical recovery windows close before referral opportunities are recognized. Relevant information is buried across disconnected documents with no systematic way to connect symptoms, barriers, and program eligibility.

DISCOVERY

How I approached discovery

USER INTERVIEWS

Healthcare Programs SMEs and case managers

Case managers need to understand why a recommendation was generated — not just what to do. Trust requires evidence traceability.

WORKFLOW OBSERVATIONS

Claim review process mapping

Manual document cross-referencing is the bottleneck — not decision-making itself. The cognitive load is in finding, not judging.

STAKEHOLDER INTERVIEWS

Data scientists, architects, program managers

Production claim data unavailable during sprint — synthetic data strategy needed before any model work could begin.

RESEARCH

InnoFest 2025 challenge framing

36 ideas submitted org-wide. Challenge: 'How can we utilize AI to automate tasks, enhance provider collaboration, and improve program measurement?'

ANALYTICS

TODO: production claim complexity metrics

Document volume per claim (20–75) and review time (90+ min) sourced from SME input. TODO: validate with operational data.

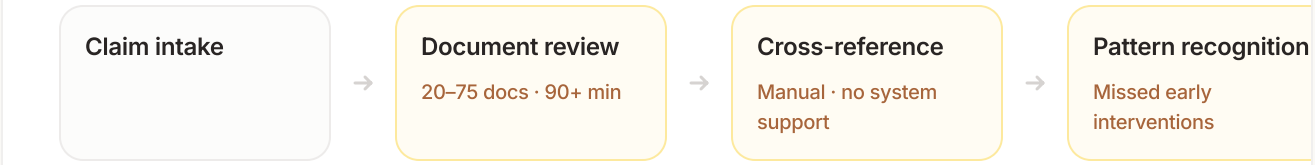
PRODUCT JUDGMENT

“We intentionally kept clinicians responsible for final referral decisions. AI surfaced recommendations, but never replaced clinical judgment.”

CURRENT WORKFLOW

Before — and where it broke

Current case manager workflow — cognitive load at every step



PRODUCT JUDGMENT

“We mapped cognitive load to the document review step — the bottleneck was finding information, not making decisions. AI should reduce search, not replace judgment.”

PRODUCT THINKING

Observation → insight → decision

DECISION CHAIN 1

OBSERVATION	INSIGHT	DECISION	TRADEOFF	EXPECTED OUTCOME
Initial concepts focused on recommendation lists — traditional AI output format.	Case managers need to explore connections between symptoms, barriers, programs, and claim events — not scan a ranked list.	Push beyond recommendation list toward interactive visual mind map.	Higher design and engineering complexity vs. faster MVP with simple list UI.	Mind map became the most memorable part of the demonstration; data architect independently reached out about the visualization approach.

DECISION CHAIN 2

OBSERVATION	INSIGHT	DECISION	TRADEOFF	EXPECTED OUTCOME
Production claim data could not be used during the innovation challenge.	Healthcare SME had limited availability — blocking the data science team's ability to train and test models.	Use ChatGPT to generate realistic claim scenarios; validate with experienced case manager before providing to data science.	Synthetic data may not capture all edge cases vs. complete blocker without any training data.	Unblocked model development and testing while respecting privacy and security requirements.

DECISION CHAIN 3

OBSERVATION	INSIGHT	DECISION	TRADEOFF	EXPECTED OUTCOME
AI could surface referral recommendations with supporting evidence.	Clinical judgment must remain with case managers — AI recommending is different from AI deciding.	AI surfaces recommendations and evidence; humans make final referral decisions.	Slightly lower automation potential vs. higher clinician trust and safer workflow.	RTW Compliance Manager asked: 'Can this be used for other subject matters such as pain?' — signaling trust in the approach.

DECISION CHAIN 4

OBSERVATION	INSIGHT	DECISION	TRADEOFF	EXPECTED OUTCOME
2-week sprint with no formal authority over cross-functional team.	MVP scope clarity and aligned success criteria matter more than feature count in constrained sprints.	Define MVP around explainability (evidence trace + mind map) rather than recommendation accuracy alone.	Narrower feature set vs. coherent demo that stakeholders could evaluate.	2nd place Process Transformation; strong engagement during final showcase.

PRODUCT JUDGMENT

“We pushed beyond a recommendation list because discovery showed case managers need to explore connections — explainability is the product, not the model.”

KEY PRODUCT DECISIONS

Decision matrix

KEY PRODUCT DECISIONS

Decision	Alternative	Why we chose this	Tradeoffs	Outcom
AI surfaces recommendations; humans decide HUMAN VS AI	Fully automated referral routing	Clinical judgment needed to remain with case managers; trust and safety require human-in-the-loop.	Slightly lower automation vs. higher trust and safer workflow	Stakeho asked a extendi other cli domains manage
Interactive mind map over recommendation list USER EXPERIENCE VS COMPLIANCE	Traditional ranked recommendation UI	Discovery showed case managers need to explore connections, not scan lists — explainability is the product.	Higher design/engineering complexity in 2-week sprint	Most me demo el data arc praised visualiz approac
Synthetic data via ChatGPT + case manager validation SPEED VS SAFETY	Wait for production data access or anonymized extracts	Production data unavailable; SME time limited — synthetic data unblocked the entire data science workstream.	Edge case coverage risk vs. complete blocker without data	Team sh working prototyp sprint constrai
Evidence traceability as core feature HUMAN VS AI	Black-box recommendations with confidence scores only	Case managers won't act on recommendations they can't verify against source documents.	Additional engineering for document linking and snippet extraction	Transpa became differen stakeho feedbac

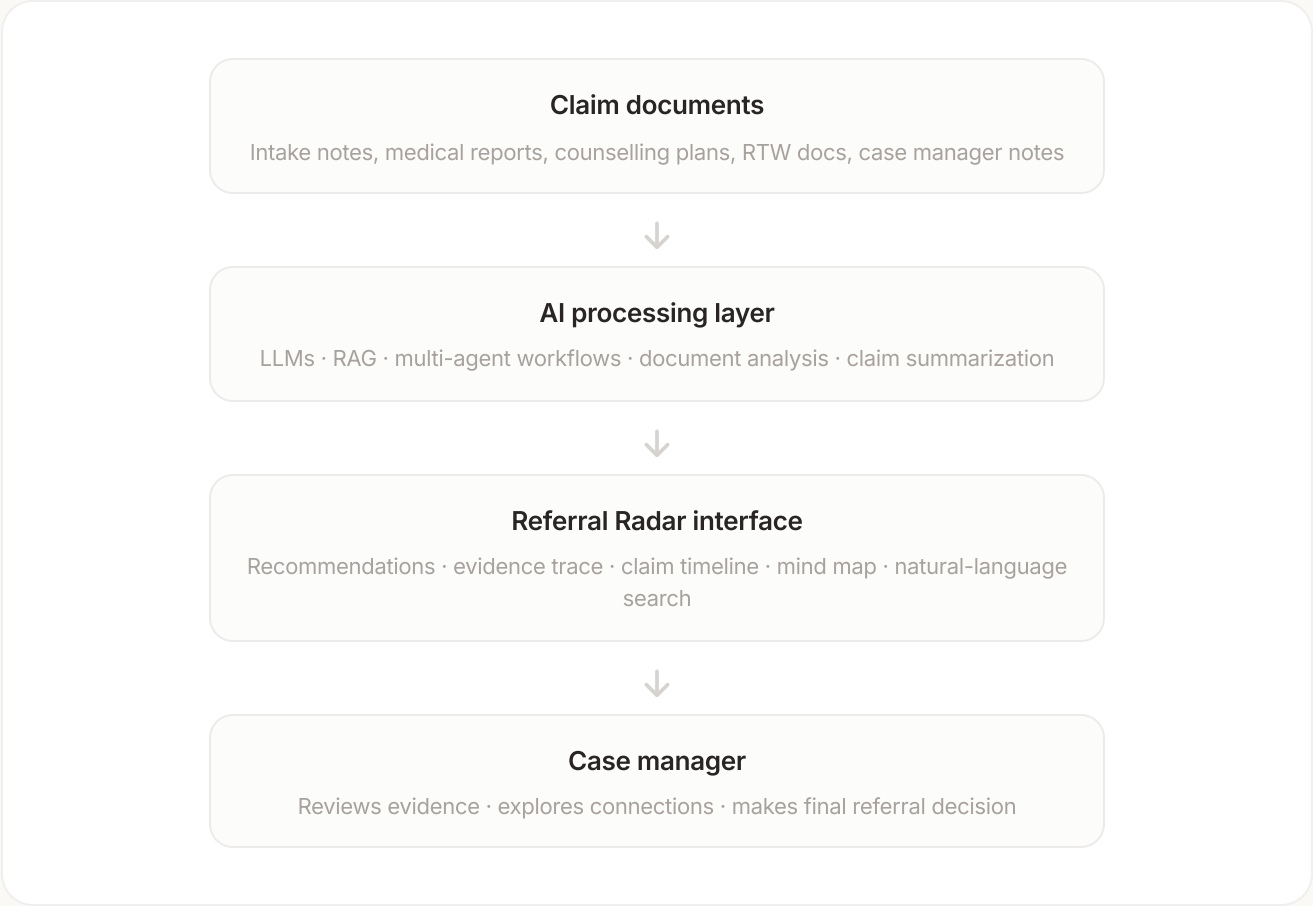
PRODUCT JUDGMENT

"We used synthetic data via ChatGPT + case manager validation because waiting for production data would have killed the sprint — unblock the team, validate realism with SMEs."

TECHNICAL COLLABORATION

Architecture at product level

Product-level architecture — AI-assisted claim intelligence



PRODUCT RESPONSIBILITIES

- MVP scope and product vision
- Synthetic data strategy and case manager validation
- Experience design (mind map, evidence trace)
- Stakeholder alignment across 2-week sprint
- Human-in-the-loop boundary definition

ENGINEERING RESPONSIBILITIES

- LLM integration and RAG pipeline
- Multi-agent workflow orchestration
- Document analysis and claim summarization
- Mind map visualization and search interface
- Agent trace view implementation

SHARED RESPONSIBILITIES

- Synthetic data quality criteria with healthcare SME
- Evidence linking accuracy validation
- Demo scenario selection and showcase preparation

TECHNICAL TRADEOFFS

Synthetic data enabled speed but TODO: production validation needed for edge cases

Multi-agent complexity vs. single-model simplicity — chose agents for explainability

Mind map UX investment vs. faster list-based MVP

TODO: specific confidence thresholds for recommendation surfacing not documented

HUMAN RESPONSIBILITIES

Final referral decisions and program authorization

Clinical judgment on injured worker readiness

Validation of synthetic training scenarios

Prioritization among competing referral opportunities

Override or dismissal of AI recommendations

AI RESPONSIBILITIES

Surface referral recommendations ranked by priority

Extract and link supporting evidence from claim documents

Generate claim timeline summaries

Map relationships between symptoms, barriers, and programs

Natural-language search across claim context

DECISION BOUNDARIES

AI never auto-approves or routes referrals without human review

Every recommendation includes traceable evidence snippets

Agent trace view shows how AI arrived at suggestions — not just outputs

TODO: specific confidence thresholds for surfacing vs. suppressing recommendations

Case manager can explore mind map without accepting any recommendation

PRODUCT JUDGMENT

“We invested in evidence traceability over black-box confidence scores because case managers won't act on recommendations they can't verify.”

EXECUTION

How we shipped

TIMELINE

InnoFest 2025: 36 ideas submitted org-wide

Shortlisted: Process Transformation finalist

2-week rapid innovation sprint

Final showcase and judging

TODO: exact sprint dates

PRIORITIZATION

Explainability (evidence trace + mind map) over recommendation accuracy

Synthetic data creation to unblock data science

Demo-ready experience over production infrastructure

Human-in-the-loop boundaries defined before model work

MVP

Referral recommendations ranked by priority

Evidence traceability with source document snippets

Claim timeline summarization

Interactive mind map visualization

Natural-language search across claim context

RISKS

Production data unavailable — synthetic data edge case coverage

2-week sprint with no formal authority

Healthcare SME limited availability

TODO: production roadmap and scaling risks

TRADEOFFS ACCEPTED

Prototype over production-ready system

Synthetic over production training data

Mind map complexity over simple list UI

Human-in-the-loop over full automation

CROSS-FUNCTIONAL PARTNERS

Healthcare Programs SMEs and case managers

Data scientists and ML engineers

Solution architects

Full-stack developers

Program Manager, Advanced Technology Solutions

PRODUCT JUDGMENT

“We scoped MVP around explainability (mind map + evidence trace) rather than model accuracy — in a 2-week sprint, a coherent demo beats a accurate black box.”

RESULTS

Outcomes

BUSINESS OUTCOMES

2nd

PLACE · PROCESS TRANSFORMATION

Among InnoFest 2025 finalists; 36 ideas submitted org-wide

OPERATIONAL OUTCOMES

2 weeks

SPRINT TO WORKING PROTOTYPE

Zero-to-one with cross-functional team and no production data

USER OUTCOMES

90+ min

MANUAL REVIEW ADDRESSED

Prototype demonstrated faster referral opportunity identification with evidence

USER OUTCOMES

Strong

STAKEHOLDER ENGAGEMENT

Data architect praised mind map; RTW Compliance Manager asked about extending to pain management

BUSINESS OUTCOMES

TODO

PRODUCTION COST IMPACT

Resume mentions ~$20M potential cost reduction — needs validation before including in materials

OPERATIONAL OUTCOMES

36

IDEAS IN COMPETITION

Referral Radar shortlisted and placed 2nd in Process Transformation category

Referral recommendations ranked by priority

Interactive mind map — exploring claim connections

Evidence traceability with source document snippets

LESSONS LEARNED

What changed how I think

What surprised me

The mind map — not the recommendation engine — generated the strongest stakeholder response. Explainability was the product, not the AI.

What I would do differently

Define confidence thresholds and human override patterns earlier — even in a prototype, decision boundaries need explicit design.

How this changed how I think as a PM

In healthcare AI, the constraint isn't model accuracy — it's trust. Product judgment means knowing where humans must remain in control, and designing the interface to make that boundary visible.

PRODUCT JUDGMENT

“We learned that in healthcare AI, trust is the constraint — not model performance. Product judgment means designing human-AI boundaries before writing prompts.”

WHY THIS MATTERS

Relevance to healthcare product leadership

Referral Radar demonstrates how to build AI-assisted clinical workflows where trust, explainability, and human judgment are designed in — not bolted on. The same product leadership applies to any healthcare AI product where care delivery decisions require evidence, transparency, and clear human-AI boundaries.

QUESTIONS THIS PROJECT FORCED ME TO ANSWER

Head of Product questions — and how I answered them

Where should humans remain in control?

Final referral decisions, program authorization, and clinical judgment on worker readiness. AI surfaces and explains — humans decide.

Is this actually a software problem?

Yes — but the software problem is information retrieval and connection, not decision automation. The workflow bottleneck is cognitive load in document review.

What evidence changed our direction?

SME input on review time (90+ min) and case manager need for evidence traceability shifted MVP from recommendation list to interactive mind map.

What tradeoffs did we accept?

Synthetic over production data; mind map complexity over simple UI; human-in-the-loop over automation; prototype over production infrastructure.

What would happen if we built nothing?

Case managers continue spending 90+ minutes per claim on manual cross-referencing; early intervention windows close as claim complexity grows.

How would AI change this approach today?

Better synthetic data generation, faster prototyping tools (Lovable, Copilot), and more mature RAG patterns — but human-in-the-loop boundaries would remain the same.

IF I DID THIS PROJECT AGAIN

Knowing what I know today

What would I change?

Define explicit confidence thresholds and suppression rules for recommendations — even in prototype, decision boundaries need design.

What would I validate earlier?

Production data access path and anonymization feasibility — understanding data constraints on day one shapes entire architecture.

What assumptions were incorrect?

Assumed recommendation accuracy would be the demo highlight. Stakeholders cared more about explainability and visual exploration.

What would I intentionally NOT build?

Automated referral routing — even with high confidence, clinical workflows require human authorization.

How would AI change my approach today?

AI-enabled prototyping (Lovable, Copilot) could compress concept-to-validation further — resume cites ~40% faster timelines. TODO: validate.

What would I prototype before implementation?

Mind map interaction with case managers using paper prototypes — validate exploration pattern before engineering investment.

HEAD OF PRODUCT VIEW

If you only read one thing — why trust this PM with a healthcare workflow?

PROBLEM

Case managers spend 90+ minutes cross-referencing 20–75 documents per claim; early referral windows close unnoticed.

JUDGMENT

Designed AI as decision support, not decision maker. Invested in explainability (mind map + evidence trace) over recommendation accuracy.

HARDEST TRADEOFF

Mind map complexity in a 2-week sprint vs. simple list UI — chose explainability because trust is the constraint in clinical AI.

TECHNICAL CHALLENGE

No production data available — built synthetic data pipeline with ChatGPT + case manager validation to unblock entire data science workstream.

HEALTHCARE COMPLEXITY

Clinical judgment, privacy constraints, multi-document claim context, and stakeholder trust across regulated healthcare programs.

BUSINESS OUTCOME

2nd place InnoFest 2025 Process Transformation; strong cross-functional engagement; validated explainable AI approach.

PRODUCT LESSON

In healthcare AI, design human-AI boundaries before writing prompts — trust is the product.